

JSW Dealer Management System - Complete Technical Blueprint

1. Project Overview

The system is a Dealer Order, Scheme, Payment, and Credit Management Platform for JSW NeoSteel products.

Main business logic:

- Dealer places order in MT
 - Scheme applies based on payment timing
 - Early payment gets discount
 - Late payment gets penalty/fine
 - System tracks dealer outstanding and scheme utilization
 - Workflow approval process controls order lifecycle
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2. Proposed System Modules

Core Modules

1. Authentication & Role Management
 2. Dealer Management
 3. Product Management
 4. Scheme Management
 5. Order Management
 6. Invoice Management
 7. Payment Management
 8. Credit/Debit Note Management
 9. Penalty Engine
 10. Notification Module
 11. Dashboard & MIS Reports
 12. Audit Logs
 13. API Integration Module
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3. User Roles

Role	Description
Super Admin	Full system access
Management	View reports and analytics
Sales Executive	Dealer/order handling
Finance Team	Payment reconciliation
Approval Manager	Order approval authority
Dealer	Create orders and view payments
Dispatch Team	Dispatch and invoice handling
Auditor	Read-only audit access

4. Role Flow

```
Dealer
↓
Sales Executive
↓
Approval Manager
↓
Finance Verification
↓
Dispatch Team
↓
Invoice Generation
↓
Payment Tracking
↓
Penalty/Discount Calculation
↓
Closure
```

5. Complete Order Flow

Step-by-Step Order Lifecycle

Step 1 - Dealer Registration

- Dealer created by admin
- Credit limit assigned
- Credit period assigned
- GST/PAN verified

Step 2 - Scheme Creation

Admin creates:

- Scheme name
- Product applicability
- Discount slabs
- Penalty slabs
- Validity period

Step 3 - Order Placement

Dealer selects:

- Product
- Quantity (MT)
- Delivery details
- Scheme

System calculates:

- Base amount
- Discount applicability
- Final rate

Step 4 - Order Approval

Approval manager:

- Approves
- Rejects
- Sends back for modification

Step 5 - Invoice Creation

After approval:

- Invoice generated
- Dispatch initiated
- Payment due date created

Step 6 - Payment Tracking

System tracks:

- Paid amount
- Pending amount
- Due days

Step 7 - Discount/Penalty Calculation

System auto-calculates:

- Discount
- Credit note
- Debit note
- Penalty

Step 8 - Closure

Order marked completed after:

- Full payment
- Penalty settlement
- Invoice reconciliation

6. Payment Calculation Flow

Business Logic

Example

Scheme:

12000/MT

Quantity:

100 MT

Base Amount:

12,00,000

Discount Slab Example

Payment Timeline	Discount
Advance Payment	₹500/MT
1-5 Days	₹400/MT
6-10 Days	₹300/MT
11-15 Days	₹200/MT
16-20 Days	₹100/MT
After 20 Days	Penalty starts

Penalty Example

Delay Days	Penalty
After 20 Days	₹50/MT/day

Formula

Discount Formula

Discount Amount = Quantity × Discount Per MT

Penalty Formula

Penalty Amount = Quantity × Penalty Per MT × Delay Days

Final Amount

Final Amount = Base Amount - Discount + Penalty

7. Approval Workflow

Order Approval Workflow

```
Draft
↓
Submitted
↓
Under Review
↓
Approved / Rejected
↓
Invoice Generated
↓
Dispatch Completed
↓
Payment Pending
↓
Payment Completed
↓
Closed
```

8. Notification Workflow

Notification Triggers

Event	Notification
Dealer Created	Welcome Email
Order Submitted	Admin Alert
Order Approved	Dealer Notification
Invoice Generated	Email + WhatsApp
Payment Due	Reminder

Event	Notification
Overdue Payment	Penalty Alert
Payment Completed	Confirmation

9. Suggested Database Structure

users

Column	Type
id	bigint
role_id	bigint
name	varchar
email	varchar
mobile_no	varchar
password	varchar
status	tinyint
created_at	timestamp
updated_at	timestamp

roles

Column	Type
id	bigint
role_name	varchar
description	text
created_at	timestamp
updated_at	timestamp

dealers

Column	Type
id	bigint
dealer_code	varchar
company_name	varchar
gst_no	varchar
pan_no	varchar
mobile_no	varchar
email	varchar
address	text
state	varchar
city	varchar
pincode	varchar
credit_limit	decimal
credit_period	int
status	tinyint
created_at	timestamp
updated_at	timestamp

product_categories

Column	Type
id	bigint
category_name	varchar
description	text
created_at	timestamp
updated_at	timestamp

products

Column	Type
id	bigint
category_id	bigint
sku_code	varchar
product_name	varchar
base_rate	decimal
unit	varchar
status	tinyint
created_at	timestamp
updated_at	timestamp

schemes

Column	Type
id	bigint
scheme_name	varchar
start_date	date
end_date	date
description	text
status	tinyint
created_at	timestamp
updated_at	timestamp

scheme_slabs

Column	Type
id	bigint

Column	Type
scheme_id	bigint
from_day	int
to_day	int
discount_per_mt	decimal
penalty_per_mt	decimal
created_at	timestamp
updated_at	timestamp

scheme_products

Column	Type
id	bigint
scheme_id	bigint
product_id	bigint
created_at	timestamp
updated_at	timestamp

orders

Column	Type
id	bigint
order_no	varchar
dealer_id	bigint
scheme_id	bigint
order_date	date
total_mt	decimal
base_amount	decimal
discount_amount	decimal

Column	Type
penalty_amount	decimal
final_amount	decimal
order_status	varchar
approval_status	varchar
approved_by	bigint
approved_at	datetime
remarks	text
created_at	timestamp
updated_at	timestamp

order_items

Column	Type
id	bigint
order_id	bigint
product_id	bigint
quantity_mt	decimal
rate_per_mt	decimal
amount	decimal
created_at	timestamp
updated_at	timestamp

invoices

Column	Type
id	bigint
invoice_no	varchar
order_id	bigint

Column	Type
invoice_date	date
invoice_amount	decimal
due_date	date
invoice_status	varchar
created_at	timestamp
updated_at	timestamp

dispatches

Column	Type
id	bigint
order_id	bigint
dispatch_no	varchar
vehicle_no	varchar
transporter_name	varchar
dispatch_date	date
status	varchar
created_at	timestamp
updated_at	timestamp

payments

Column	Type
id	bigint
dealer_id	bigint
order_id	bigint
invoice_id	bigint
payment_date	date

Column	Type
payment_mode	varchar
transaction_no	varchar
amount	decimal
remarks	text
created_at	timestamp
updated_at	timestamp

penalties

Column	Type
id	bigint
order_id	bigint
penalty_days	int
penalty_rate	decimal
penalty_amount	decimal
created_at	timestamp
updated_at	timestamp

credit_notes

Column	Type
id	bigint
dealer_id	bigint
order_id	bigint
amount	decimal
reason	text
created_at	timestamp
updated_at	timestamp

debit_notes

Column	Type
id	bigint
dealer_id	bigint
order_id	bigint
amount	decimal
reason	text
created_at	timestamp
updated_at	timestamp

notifications

Column	Type
id	bigint
dealer_id	bigint
type	varchar
message	text
sent_status	tinyint
sent_at	datetime
created_at	timestamp
updated_at	timestamp

audit_logs

Column	Type
id	bigint
user_id	bigint

Column	Type
module	varchar
action	varchar
old_data	longtext
new_data	longtext
ip_address	varchar
created_at	timestamp

10. ER Diagram (Text Representation)

```

roles
  ↓
users

users
  ↓
orders

orders
  ↓
order_items

orders
  ↓
invoices

orders
  ↓
payments

orders
  ↓
penalties

orders
  ↓
credit_notes

orders
  ↓

```

```
debit_notes  
  
schemes  
↓  
scheme_slabs  
  
products  
↓  
order_items  
  
products  
↓  
scheme_products
```

11. Dashboard Suggestions

Management Dashboard

- Total Sales
- Outstanding Amount
- Overdue Payments
- Active Schemes
- Dealer Performance
- Monthly Revenue

Dealer Dashboard

- Active Orders
- Invoice Status
- Outstanding Amount
- Available Schemes
- Payment Due Alerts

Finance Dashboard

- Pending Payments
 - Reconciliation Status
 - Penalty Reports
 - Credit Exposure
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12. Suggested Technologies

Component	Suggestion
Backend	Laravel 7/10
Database	MySQL
Queue	Redis / Database Queue
Frontend	Blade + jQuery / Vue
Excel Export	Laravel Excel
Notification	WhatsApp API + Mail
Authentication	Laravel Auth/Sanctum
Hosting	Ubuntu + Nginx
Queue Manager	Supervisor / PM2

13. Important Development Considerations

Performance

- Use queue for reports
- Use chunk processing
- Add indexing
- Use pagination
- Cache dashboard queries

Security

- Role-based permission
- Audit logs
- Input validation
- Rate limiting
- CSRF protection

Scalability

- Queue architecture
- API-ready structure
- Modular development
- Service layer implementation

14. Important Client Clarifications Required

Before Development

1. Exact approval hierarchy
2. Final discount slab logic
3. Exact penalty formula
4. Dealer login flow
5. Dispatch integration requirement
6. Tally integration scope
7. WhatsApp provider details
8. Report formats
9. GST/tax calculation rules
10. Multi-company support required or not
11. Mobile app needed or not
12. Offline mode required or not
13. Multi-warehouse support needed or not
14. Partial payment handling logic
15. Partial dispatch handling logic

15. Recommended Development Phases

Phase 1

- Authentication
- Role Management
- Dealer Master
- Product Master
- Scheme Master

Phase 2

- Order Management
- Approval Workflow
- Invoice System

Phase 3

- Payment Management
- Penalty Engine
- Credit/Debit Notes

Phase 4

- Notifications
- MIS Reports
- Dashboard

Phase 5

- Tally Integration
- WhatsApp Integration
- Optimization
- Security Audit

16. Final Recommendation

This project has heavy business logic.

Most critical areas:

1. Discount/Penalty Engine
2. Approval Workflow
3. Payment Reconciliation
4. Reporting Performance
5. Audit Logging

Strong recommendation:

- Design database carefully before coding
- Freeze scheme logic before implementation
- Use queue for heavy operations
- Maintain audit logs for all financial actions
- Keep calculation engine modular